

1. Introduction

Background
Technology is reshaping how people communicate, work, and access services globally. Mobile technologies have improved access to information and reduced transaction costs across Sub-Saharan Africa (Aker & Mbiti, 2010). In many African cities, digital platforms have transformed urban transportation by connecting passengers and drivers via digital interfaces. Ride-hailing platforms enable users to locate drivers, track rides via GPS, and complete transactions digitally.

Context in Niger
Mobile expansion in Niger has been significant. According to GSMA(Global Association that manages the System for mobile phone) Intelligence, Niger had approximately 14.6 million mobile connections in 2023 (54.7% of population). By 2025, this rose to ~16.5 million connections (~60% of population). Mobile cellular subscriptions reached ~65.7 per 100 inhabitants in 2023 (ITU, 2024).

Problem
Despite this mobile expansion, essential urban services transportation, delivery, and local market transactions remain largely informal and manually coordinated. Urban residents depend on street taxis, moto-taxis, and informal delivery lacking transparency, real-time coordination, and pricing consistency. This study examines this gap and explores the potential of an integrated multi-service digital platform.

2. Aim and Objectives

Aim
Examine access patterns to urban transportation, delivery, and local market services in Niger (2022–2026) and explore how an integrated multi-service digital platform could improve service efficiency and accessibility.

Objectives

- Examine how urban residents access delivery, transportation, and market services between 2022 and 2026.
- Assess the role of mobile phones in shaping access to these services within the study period.
- Identify challenges with existing informal systems and explore user requirements for an integrated multi-service platform in Niger.
- Propose a unified system integrating transportation, delivery, and local market services into one centralized digital platform.
- Expected Contribution**
- The study provides an evidence-based conceptual foundation for the design of a locally relevant digital solution tailored to Niger's socioeconomic and infrastructural conditions.

3. Research Questions

Main Research Question
How can an integrated digital platform improve access to urban transportation, delivery, and market services in Niger?

Sub-Questions

- How have urban residents in Niger accessed transportation, delivery, and local market services between 2022 and 2026?
- What challenges do users face with existing informal service systems, and what features and requirements are necessary for an effective multi-service digital platform in the Nigerien context?
- How has mobile technology adoption influenced access to these services?
- How could an integrated digital platform improve efficiency, coordination, transparency, and accessibility of urban services in Niger?

These questions guide the study toward a comprehensive understanding of both the current informal service landscape and the potential for digital transformation in Niger.

4. Assumptions & Significance

Key Assumptions
Mobile phone access, internet connectivity, and user trust are sufficient preconditions for platform adoption. Urban residents will benefit from digitally coordinated services, and an integrated platform can address the fragmentation of informal service systems. Digital literacy and infrastructure support are assumed to be present at a level sufficient for basic platform use among the target population.

Significance

- Academic:** Expands ICT4D research to under-studied Sahelian contexts in West Africa.
- Practical:** Provides insights for technology developers and entrepreneurs designing digital solutions for Niger.
- Policy:** Informs policymakers seeking to promote digital innovation, economic inclusion, and improved urban service coordination.
- Social:** Contributes to reducing service inefficiencies for informal workers, small business owners, and daily commuters.

Investments in digital infrastructure in Sub-Saharan Africa are associated with improved service delivery and increased participation in the digital economy (Advancing Digital Economy in Africa, 2023).

5. Theoretical Foundations

TAM

Technology Acceptance Model — Davis (1989)
Explains adoption through perceived usefulness (belief that technology improves performance) and perceived ease of use (belief that technology is easy to learn and operate). When users find a platform both useful and easy to use, adoption likelihood increases significantly. TAM is widely applied in information systems research but its application to informal service environments in developing countries remains limited.

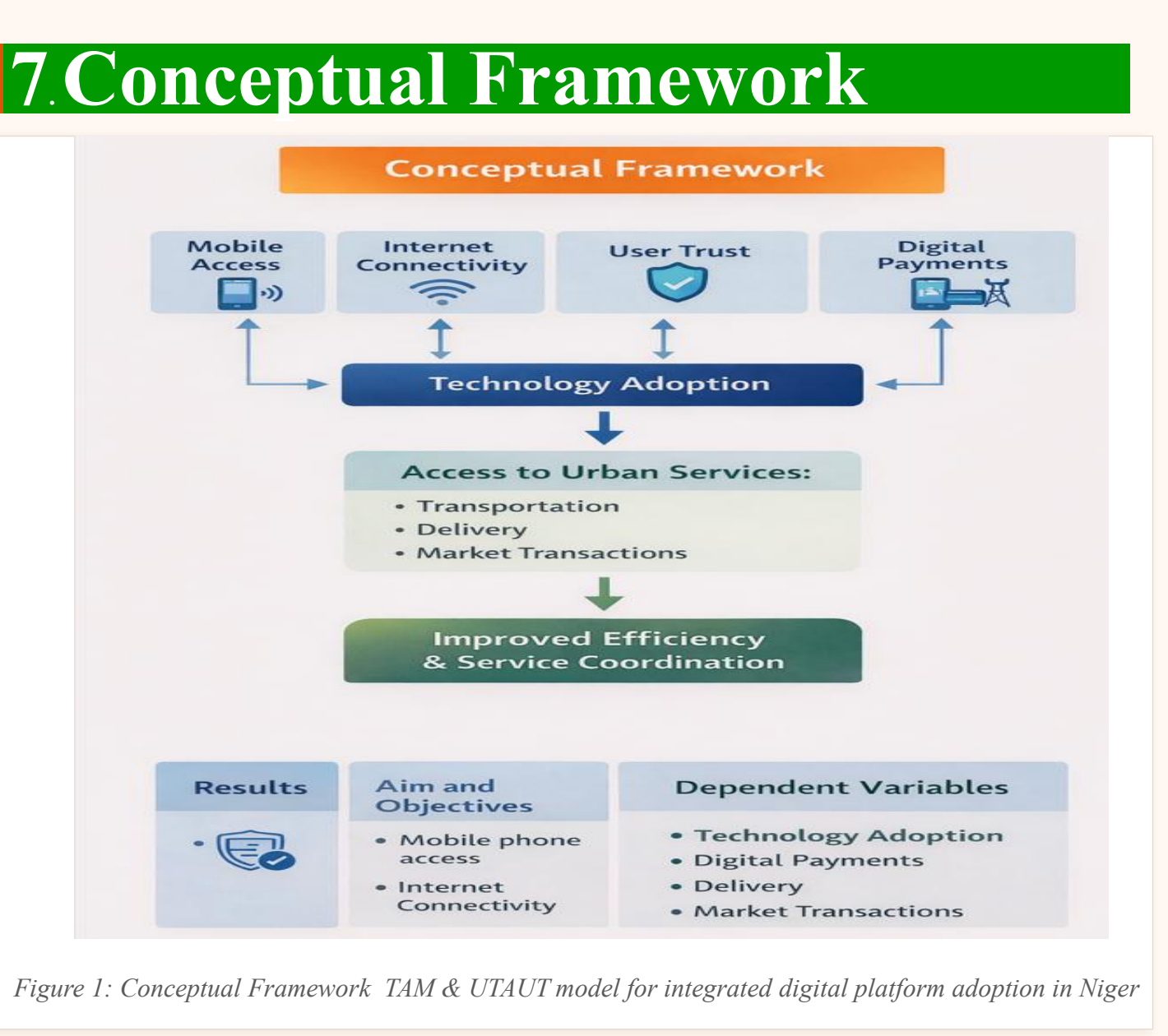
UTAUT

Unified Theory of Acceptance and Use of Technology(UTAUT) — Venkatesh et al. (2003)
Integrates multiple adoption models into four key determinants: performance expectancy, effort expectancy, social influence, and facilitating conditions. Particularly relevant for this study as it accounts for external infrastructure and social influence factors critical in low-income urban environments such as Niamey. UTAUT has been widely applied to mobile technology adoption in developing economies.

6.Research Gaps

Gap in Previous Studies	Focus of This Study	Why It Matters
Research focused on Ghana, Kenya, Nigeria only	Niger-specific urban context	Captures unique Sahelian conditions
Single services examined (ride-hailing, fintech)	Integrated multi-service platform	Holistic digital inclusion
TAM/UTAUT rarely applied to informal economies	Apply frameworks to Niger informal context	Broadens theory applicability
Limited research on integrated multi-service platforms	Propose unified digital platform for Niger	Fills critical innovation gap

These gaps highlight the need for context-specific, integrated digital solutions within emerging digital economies. Niger presents a unique case where rapid mobile growth coexists with deeply informal service structures, creating both opportunity and urgency for platform innovation.



8. Research Paradigm & Approach

Research Approach — Quantitative
The study adopts a quantitative research approach to: measure relationships between variables; test adoption-related constructs; and identify statistical patterns in user behavior. Quantitative methods allow objective evaluation of how technological and infrastructural factors influence willingness to adopt a multi-service platform in Niger.

9. Research Context & Design

Study Area
The study focuses on urban areas of Niger, particularly Niamey — the capital and largest city. Niamey is selected because it has the highest concentration of mobile users, represents the center of informal transport and market activity, and is the most likely environment for early digital platform adoption.

Research Design
A descriptive and explanatory research design is used.

- Descriptive: Describes current access patterns to transportation, delivery, and market services in Niger between 2022 and 2026.
- Explanatory: Examines how independent variables (mobile access, trust, infrastructure) influence technology adoption intention through structured quantitative surveys.

10. Reasoning

Deductive Reasoning
Deductive reasoning guides this study. Established theories (TAM and UTAUT) provide the hypotheses that are tested through empirical data collection. This top-down approach starts from existing theoretical frameworks and moves toward specific observations and measurements in the Nigerien urban context.

Reasoning Chain

- TAM/UTAUT constructs define measurable variables (perceived usefulness, effort expectancy, social influence, facilitating conditions).
- Survey instrument operationalizes these constructs into Likert-scale questionnaire items.
- Statistical analysis tests relationships among variables.
- Evidence-based conclusions are drawn about digital platform adoption feasibility in Niger.

11. Philosophy

Positivism
Positivism underpins this research. Knowledge is grounded in observable evidence and measurable data. The study assumes that:

- Social phenomena (service access, technology adoption) can be measured objectively through structured questionnaires.
- Reliable statistical patterns can be identified across the sampled population.
- Findings can support generalizations to similar urban contexts in developing economies.
- Research can be value-free and replicable.

Alignment with Theory

- Positivism aligns with both TAM and UTAUT: both frameworks use quantifiable constructs such as perceived usefulness, effort expectancy, performance expectancy, and facilitating conditions to explain and predict technology adoption behavior. This philosophical alignment ensures internal consistency throughout the study.

12. Data Collection & Analysis

01

Target Population: Urban commuters, informal transport users (taxi and moto-taxi users), market traders, small business owners, and delivery service users in Niamey, Niger. These groups represent both current users of informal urban services and potential adopters of digital platforms.

02

Sampling Strategy: Convenience sampling due to accessibility constraints and the informal structure of the population. Although non-probability sampling limits generalization, it is appropriate for exploratory digital adoption studies in emerging economies. Estimated sample size: 80–120 respondents.

03

Primary Data — Structured Questionnaire: 5 sections: (A) Demographics; (B) Urban service usage patterns; (C) Mobile and internet access; (D) TAM and UTAUT adoption factors including perceived usefulness, effort expectancy, social influence, and facilitating conditions; (E) Platform adoption intention. Likert scale (1–5).

04

Secondary Data Sources: Academic journals; World Bank and ITU reports; digital economy datasets from DataReportal and GSMA Intelligence for contextual and theoretical support on Niger's digital landscape.

05

Validity & Reliability: Content validity ensured by aligning questionnaire items with research objectives and TAM/UTAUT constructs from Chapter 2. Reliability maintained through consistent question wording, standardized response scales, and conceptual pilot testing.

06

Data Analysis Techniques: Descriptive statistics: frequencies, percentages, mean scores for Likert-scale items. Visual tools: tables and charts. Inferential analysis: regression-based methods to identify adoption patterns, digital readiness levels, and key barriers to platform adoption.

13. Ethical Considerations

Ethical Standards

- Participation is fully voluntary.
- Informed consent is obtained from all respondents before data collection begins, with clear explanation of the study's aims, procedures, and confidentiality measures.
- Responses are completely anonymous no personal identifiers are collected or stored.
- Data is used strictly for academic purposes only.
- Participants retain the right to withdraw at any stage without consequence or penalty.
- Data will be securely stored and handled in full compliance with Ashesi University ethical guidelines and research integrity standards.

14. Conclusions

Summary
This study investigates how access to urban transportation, delivery, and market services has evolved in Niger between 2022 and 2026, and whether an integrated multi-service digital platform can address existing inefficiencies. Guided by TAM and UTAUT, a quantitative positivist methodology is used to measure adoption readiness and identify barriers.

Expected Findings
Mobile phone access, user trust, and infrastructure support are expected to be the strongest predictors of platform adoption intention. The study will identify specific barriers such as digital literacy gaps, infrastructure limitations, and trust deficits that need to be addressed for successful platform deployment.

Impact
Results will inform technology designers, policymakers, and digital entrepreneurs seeking context-specific solutions for Niger's urban service challenges, contributing to the broader ICT4D literature on platform-based service delivery in Sahelian Africa.

15. Key References

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